Off the Prior: Mapping Where a Synthetic-Prior Time-Series Foundation Model Fails, and Why

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Abstract

Timestral (Mistral, October 2026; 5M to 209M parameters) is a time-series foundation model pretrained mostly on synthetic trend + season + noise series. We map where its forecasts fall behind the best achievable forecast, using simulators whose true state is known: a Monte Carlo continuation from that state serves as an oracle, so a large error counts as a failure only when the oracle does better. On series like the pretraining data, all four sizes stay at 0.99–1.15× the oracle's MASE up to horizon 128; heavy tails, long memory, bursts and multifractal cascades stay at 0.94–1.34×. The gaps are on smooth, nearly deterministic dynamics (Mackey-Glass at $\tau = 30$: 21.87× at 5M, 13.68× at 209M; for Lorenz the gap is confined to the first Lyapunov times) and on recent regime switches (3.77× at 5M and 2.43× at 209M sixteen steps after a level jump). Linear readouts of the forecast tokens recover the new level after a jump with R^2 0.956–0.980 while the forecast reaches 0.570–0.726: the change is represented but not used. A nearest-neighbour forecaster built from the context beats every size on the delay and smooth-chaos systems, and a training-free switch that backtests the two on the context lowers chaos errors to 0.824× (5M) and 0.982× (209M) of Timestral's on fresh samples, with in-prior and regime series unchanged and tails at most 1.003×. We release two benchmarks with oracle quantiles (372 tagged diagnostic series; 3,776 leaderboard series), on which Timestral-209M is at 2.00× and 1.31× the oracle's MASE.

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1. Introduction

Timestral is a family of open time-series foundation models (5M, 28M, 123M and 209M parameters) released by Mistral on October 15, 2026. It forecasts up to 1,024 steps per forward pass from up to 32,768 steps of history, outputs 21 quantiles per step, accepts past-only and known-future covariates, and was pretrained mostly on synthetic data: simple trend, seasonality and noise processes combined into multivariate series. Mistral reports first place among pretrained and zero-shot models on GIFT-Eval, TIME and fev-bench.

A model trained on a synthetic distribution learns a prior over what series look like. Benchmarks of real data measure how well that prior matches the series people happen to collect. They do not say where the prior ends. This paper asks that question directly: we build parameterized generators whose difficulty can be turned up smoothly, forecast them with all four Timestral sizes, and measure the gap to forecasters that know the true generating process. Then we look inside the model to explain the gaps, package the hard cases as benchmarks, and test whether cheap fine-tuning closes them.

Contributions:

1. **A failure map** of Timestral across four signal families (chaos and dynamics, regime switches, heavy tails and fractal processes, and a learned mixture of primitives), along every difficulty knob, at four model sizes, four context lengths and horizons from 16 steps to rollouts past one 1,024-step block, measured against a true-dynamics oracle and classical baselines.
2. **A covariate test** of Timestral's one-way covariate attention on coupled systems.
3. **A comparison of search methods** (CMA-ES, MAP-Elites quality-diversity, and gradient search through the model) for finding failures automatically.

4. **A mechanism analysis** with attention measurements on the forecast tokens and linear readouts of internal states.
5. **Two benchmarks** (a tagged diagnostic set and a leaderboard set) with generator code, seeds and a one-command evaluation.
6. **A fine-tuning study** comparing full fine-tuning, LoRA and data mixing under leave-one-family-out and held-out-knob protocols, with a regression check on normal series.

2. Setup

2.1 The model

Timestral is a causal patch transformer. A series is cut into patches of 32 steps; each patch is normalized with the mean and standard deviation of everything observed up to the end of that patch, squashed with `asinh`, and embedded. The sequence holds the context patches followed by learned forecast tokens ("Decoupled Horizon Forecasting"): each forecast token reads the whole context and the earlier forecast tokens, and a head maps it to 21 non-crossing quantiles for each of its 32 steps. Layers alternate attention over time with attention across the variates of a series. Covariate rows are read by the main-series rows but never read them. One forward pass covers up to 32 forecast patches (1,024 steps); longer horizons are produced block by block, feeding each block's median back as context. The released forecaster also averages the forecast of the series with the reflected forecast of its negation ("symmetric averaging") and clips at zero when the context is non-negative. We use the released code and weights with these defaults throughout.

Because a forecast block is causal across its forecast tokens, a forecast's first `h` steps do not depend on the requested horizon beyond `h`. We checked that slicing a 1,024-step forecast reproduces separate shorter forecasts to float precision (`tests/test_harness.py`), and evaluate all horizons up to the longest from one call.

2.2 References: what counts as a failure

A forecast error is only a failure relative to something that could have done better. We use two references:

- **True-dynamics oracle.** Every generator is a simulator with an explicit state. To build a test case we simulate a warmup and the context, copy the final state 501 times, and advance every copy over the horizon with fresh noise. Copy 0 is the realized future; the other 500 form the oracle's predictive distribution, from which we take the same 21 quantiles Timestral outputs. The oracle knows the equations, the parameters and the latent state at the forecast origin, so it is the best achievable forecast given that state (up to Monte Carlo error) and slightly stronger than anything that sees only the noisy observations. Deterministic systems get small dynamic noise and observation noise so the oracle's spread grows with the horizon instead of being zero.
- **Classical models.** Seasonal naive, ETS (additive error, damped trend, additive season when the period allows) and ARIMA (small order grid chosen by AIC), each turned into the same 21 quantiles from its Gaussian forecast distribution, fitted on the last 1,024 steps.

For every series we report the ratio of Timestral's error to the oracle's and to the best classical model on that series, and the gap (difference) to the oracle.

2.3 Metrics

- **MASE**: mean absolute error of the median, divided by the in-sample mean absolute error of the seasonal naive forecast on the context.
- **sCRPS**: the continuous ranked probability score approximated by the average pinball loss over the 21 quantile levels (times 2), divided by the same scale as MASE.
- **Coverage and calibration**: the fraction of realized values inside the 80% and 98% intervals, and the mean absolute difference between each quantile level and the fraction of realized values below that quantile.
- **WQL** (GIFT-Eval's weighted quantile loss on levels 0.1 to 0.9), reported for the leaderboard only, since it divides by the series' magnitude and breaks near zero.

Ratios are aggregated over seeds by geometric mean, with 95% bootstrap intervals.

3. Signal families

Each family is a set of simulators with a difficulty knob (full definitions in `otp/generators/`):

- **Chaos and dynamics**: Lorenz-63 (knob ρ through the onset of chaos; also the sampling interval), Rössler (knob c through period doubling), Mackey-Glass (knob: delay τ), the logistic map (knob r through the onset of chaos at about 3.57), and Kuramoto coupled oscillators (knob: coupling K through the synchronization threshold).
- **Regime switches**: jumps in mean, variance or seasonal frequency, placed inside the context at a chosen distance before the forecast origin or occurring in the horizon at a known hazard rate; and a Markov-switching AR process (knob: switching probability).
- **Heavy tails and fractal**: AR(1) and random walks with α-stable innovations (knob α), fractional Brownian motion (knob: Hurst exponent), a self-exciting (Hawkes) count process (knob: branching ratio), and a Markov-switching multifractal cascade (knob: number of multipliers).
- **Learned mixture of primitives**: a genome of four slots, each choosing one component (trend, season, AR noise, or any generator above), a weight, an additive or multiplicative role and three parameters, plus a global noise level (`otp/generators/mixture.py`). Search methods evolve genomes to maximize Timestral's error relative to the oracle while requiring the oracle to beat seasonal naive by a margin, so the series stays forecastable.
- **Controls**: in-prior series (random trend + season + AR noise, the kind of data Timestral was pretrained on).

4. Failure map

All ratios below are the geometric mean over seeds of Timestral's MASE divided by the oracle's MASE on the same series, context and horizon (1.0 = as good as the true dynamics). "vs best classical" divides by the best of seasonal naive, ETS and ARIMA on each series instead. Unless stated otherwise: context 2,048 steps, horizon 128 steps, 8 seeds per setting.

4.1 Control: series like the training data

On random trend + season + AR-noise series (the kind Timestral was pretrained on), every size stays within 0.99–1.15 of the oracle at horizons up to 128 steps. The gap grows at long horizons:

size	H=16	H=128	H=1024	H=2048
5M	1.02	1.06	1.42	2.15
28M	0.99	1.03	1.30	2.25
123M	0.99	1.01	1.12	1.36
209M	1.00	1.02	1.16	1.57

Part of that long-horizon gap is not a prior failure: the oracle knows the exact trend slope and seasonal shape, while any forecaster must estimate them from the context, and estimation error in a slope grows linearly with the horizon. Family results are read against this control.

4.2 Chaos and dynamics

series	5M	28M	123M	209M	209M vs best classical
lorenz63/rho20	1.07	1.04	1.02	1.00	0.93

series	5M	28M	123M	209M	209M vs best classical
lorenz63/rho28	1.31	1.28	1.27	1.24	0.95
lorenz63/rho28_oe2	29.89	30.89	27.49	27.91	0.75
rossler/c3	1.53	1.26	1.15	1.17	0.22
rossler/c5.7	3.31	2.61	2.42	2.04	0.25
rossler/c9	7.73	6.30	5.45	4.30	0.45
mackey_glass/tau10	1.47	1.32	1.22	1.16	0.11
mackey_glass/tau17	9.57	6.31	6.80	6.67	0.35
mackey_glass/tau30	21.87	21.38	12.88	13.68	0.48
logistic_map/r3.5	1.36	1.32	1.17	1.17	0.05
logistic_map/r3.9	1.05	1.04	1.04	1.03	0.99
kuramoto/K1	12.74	10.58	9.89	9.45	0.36
kuramoto/K8	5.18	2.28	1.86	1.43	0.11

Timestral beats the best classical model on most of these systems (the 209M model's median ratio to the best classical model across the chaos configs is 0.36) and is still far from the oracle on smooth, nearly deterministic oscillators: Mackey-Glass with long delays (21.87 at 5M, 13.68 at 209M for $\tau = 30$), weakly coupled Kuramoto oscillators (12.74 and 9.45 at $K = 1$), and the Rössler system (7.73 and 4.30 at $c = 9$). Size matters for some of these and not others: strongly coupled Kuramoto goes from 5.18 at 5M to 1.43 at 209M, while the densely sampled Lorenz system stays at 29.89 (5M) and 27.91 (209M). The chaotic logistic map ($r = 3.9$), whose predictability is gone after a few steps, stays near 1 at every size.

The gap lives inside the predictability horizon. A Lyapunov time is how long it takes a small error in a chaotic system to grow by a factor of e (for Lorenz at $\rho = 28$ the exponent is 0.907 per time unit). Putting four sampling intervals of the same Lorenz system on that common axis shows that the ratio is set by how many Lyapunov times the forecast reaches, not by the step count:

sampling step (time units)	horizon (steps)	horizon (Lyapunov times)	209M MASE / oracle
0.02	16	0.29	6.88
0.05	16	0.73	11.32
0.10	16	1.45	27.32
0.02	128	2.32	27.91
0.20	16	2.90	15.87
0.05	128	5.81	2.65
0.10	128	11.61	1.24
0.02	1024	18.58	1.22
0.20	128	23.22	1.18
0.02	2048	37.15	1.10
0.05	1024	46.44	1.09
0.10	1024	92.88	1.02
0.05	2048	92.88	1.06
0.10	2048	185.76	1.01
0.20	1024	185.76	1.02
0.20	2048	371.52	1.01

Inside one to three Lyapunov times the oracle can still track the trajectory and Timestral cannot; past about ten, nobody can, and the ratio returns to about 1. The noise level of the generator sets how good the oracle can be inside that window. Sweeping the noise level (dynamic and observation noise together) shows how much of each ratio that choice sets:

system	noise (% of scale)	5M	28M	123M	209M
lorenz63_rho28	0.10	1.96	1.89	1.87	1.83

system	noise (% of scale)	5M	28M	123M	209M
lorenz63_rho28	0.30	1.65	1.60	1.57	1.55
lorenz63_rho28	1.00	1.48	1.47	1.44	1.44
lorenz63_rho28	3.00	1.24	1.21	1.19	1.18
lorenz63_rho28	10.00	1.11	1.09	1.07	1.07
mackey_glass_tau17	0.10	75.47	64.81	56.19	43.93
mackey_glass_tau17	0.30	22.32	18.29	12.69	12.68
mackey_glass_tau17	1.00	6.32	5.75	4.84	4.32
mackey_glass_tau17	3.00	2.71	2.60	2.25	2.14
mackey_glass_tau17	10.00	1.76	1.81	1.48	1.46
rossler_c5.7	0.10	17.33	15.70	12.84	11.77
rossler_c5.7	0.30	10.57	9.80	7.90	6.78
rossler_c5.7	1.00	2.51	2.32	2.12	1.97

At 0.1% noise the oracle is nearly exact and Mackey-Glass sits at 43.93–75.47 times its error; at 10% the gap is 1.46–1.81. The ratio is therefore a measure of how much predictable structure Timestral leaves unused, and its size depends on how much predictable structure the generator provides. The ordering of model sizes holds at every noise level.

4.3 Heavy tails and fractal processes

Across α-stable AR processes and random walks, fractional Brownian motion, Hawkes counts and multifractal cascades, all four sizes stay between 0.94 and 1.34 of the oracle:

series	5M	28M	123M	209M	209M vs best classical
tails/stable_ar/alpha2_rho0.9	1.00	1.01	1.00	1.00	1.03
tails/stable_ar/alpha1.3_rho0.9	0.99	0.99	1.00	1.01	0.97
tails/stable_ar/alpha1.3_rho1.0	1.08	1.34	1.09	1.13	1.14
tails/fbm/H0.2	0.99	0.99	0.98	0.99	1.00
tails/fbm/H0.8	1.17	1.20	1.26	1.15	0.76
tails/hawkes/eta0.3	1.00	1.01	1.00	1.00	1.00
tails/hawkes/eta0.95	1.06	1.03	1.03	1.04	1.09
tails/msm/k8_ret	1.00	1.00	1.00	1.00	1.00
tails/msm/k8_cum	1.03	1.07	1.02	1.00	1.04

Heavy tails and long memory, which a trend + season + Gaussian-noise description of series does not cover, do not produce failures of the size seen for deterministic dynamics.

4.4 Regime switches

Three kinds of switch (a level jump, a change of seasonal period, a change of noise level) placed a chosen number of steps before the forecast origin, so the context shows the switch:

switch	size	16 steps ago	64 steps ago	256 steps ago	1024 steps ago
level +4 sd	5M	3.77	1.64	1.09	1.06
level +4 sd	209M	2.43	1.06	1.02	1.02
period x2	5M	2.43	2.14	1.22	1.07
period x2	209M	2.38	1.84	1.10	1.04
noise sd x0.25	5M	1.35	1.41	1.16	1.11
noise sd x0.25	209M	1.06	1.16	1.05	1.03

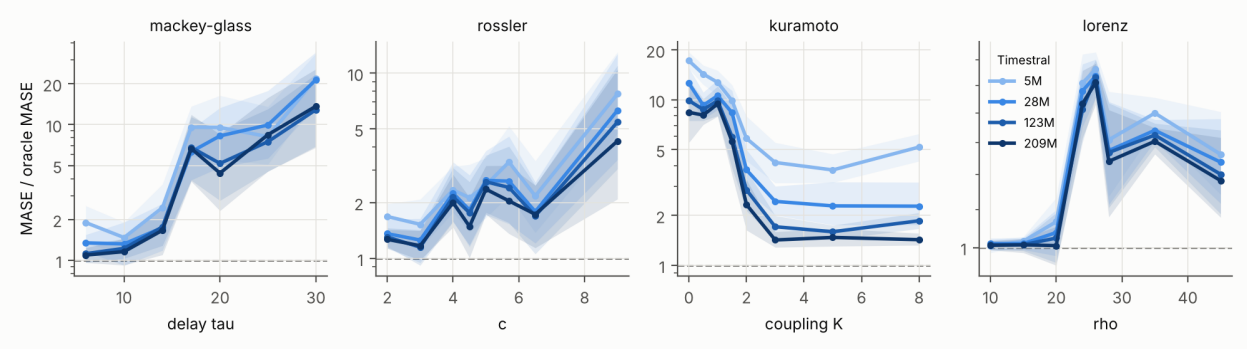


Figure 1: MASE ratio to the oracle along each chaos knob (context 2,048, horizon 128; shaded: 95% bootstrap interval over 8 seeds; dashed: as good as the oracle).

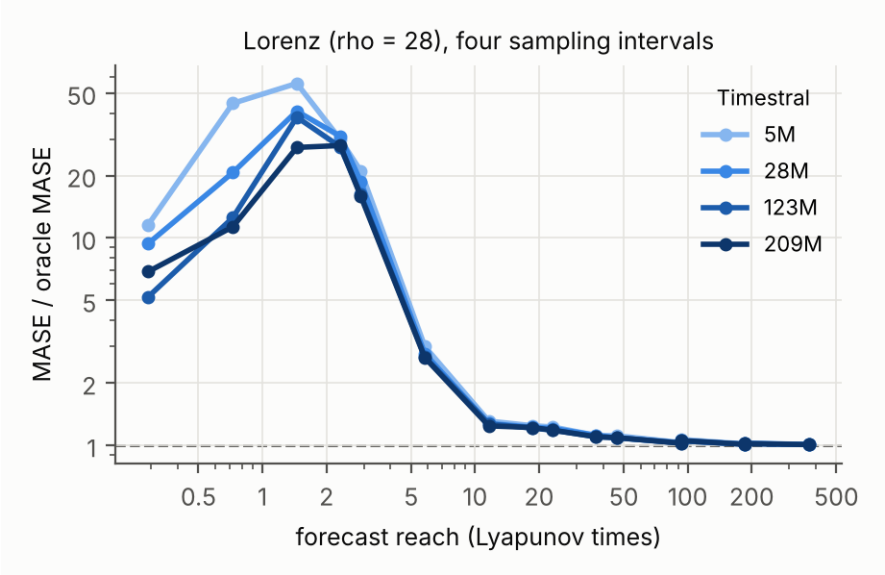


Figure 2: Lorenz at four sampling intervals on one axis: forecast reach in Lyapunov times.

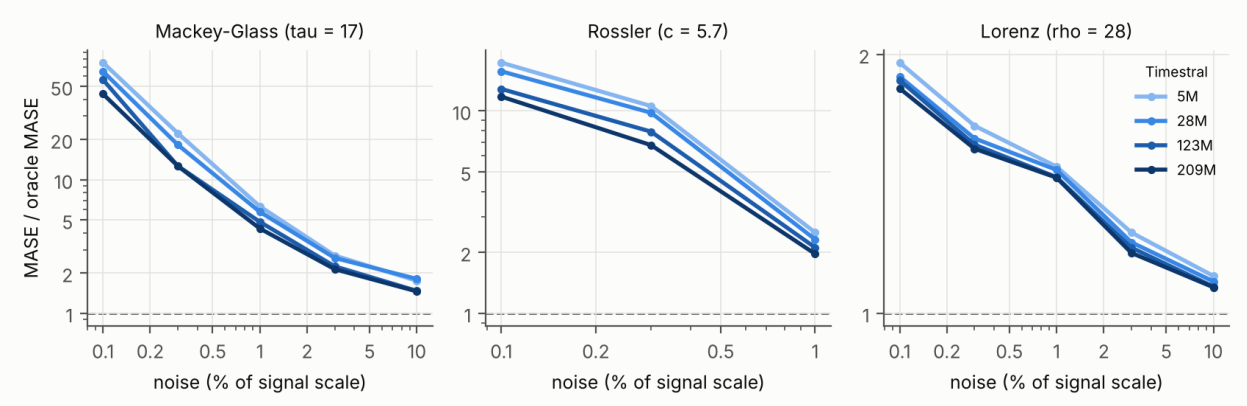


Figure 3: Ratio to the oracle against the noise level of the generator (16 seeds, horizon 128).

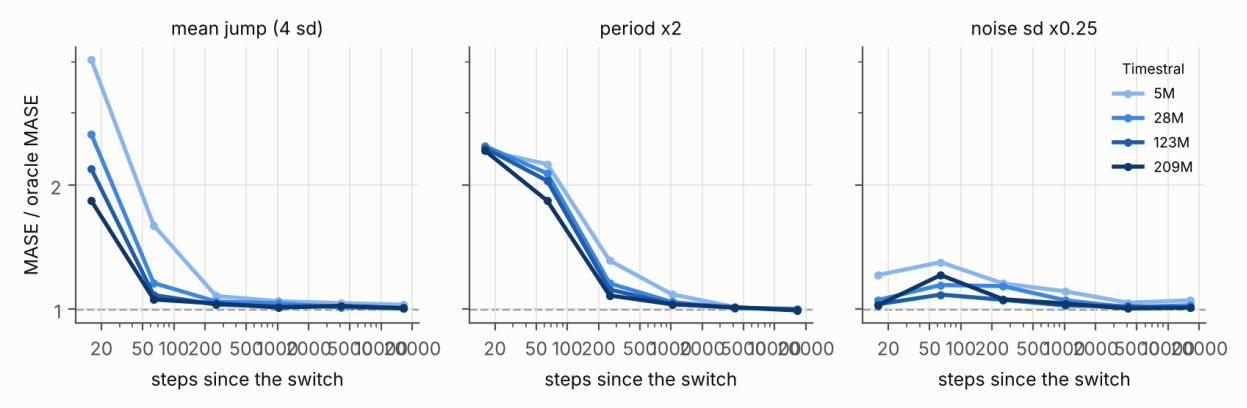


Figure 4: Ratio to the oracle against the number of steps since the switch (context 8,192, horizon 128).

The failure is concentrated right after a switch. Sixteen steps after a level jump of 4 noise standard deviations, the error is 3.77 times the oracle's at 5M and 2.43 at 209M; a thousand steps later it is 1.06 and 1.02. Period changes last longer: 64 steps after the period triples, the ratio is still 2.24 (5M) and 2.12 (209M). A direct measure of how much of a level jump the forecast adopts (1 = forecasts at the new level, like the oracle; 0 = still at the old level), over the first 16 forecast steps and pooled over jump sizes, is 0.43 / 0.66 / 0.96 at 16 / 64 / 256 steps after the switch for 5M and 0.49 / 0.70 / 0.93 for 209M. Section 7.3 shows that the change is nevertheless represented inside the model.

When the switch happens during the horizon at a known rate (so nobody can see it coming), Timestral stays at 0.93–1.06 of the oracle, and on Markov-switching AR processes at 1.01–1.13.

4.5 Does more history help?

All four sizes with contexts up to 32,768 steps (Colab job B, 28 settings × 8 seeds):

series	size	T=2048	T=8192	T=32768
mackey_glass/tau17	5M	9.09	6.91	6.47
mackey_glass/tau17	209M	4.91	4.54	3.84
kuramoto/K1	5M	15.26	13.86	13.74
kuramoto/K1	209M	9.47	8.10	7.62
rossler/c5.7	5M	2.83	2.59	2.49
rossler/c5.7	209M	2.23	2.10	2.06
lorenz63/rho28	5M	1.37	1.32	1.30
lorenz63/rho28	209M	1.25	1.21	1.19
regime/mean_jump/context/delta4/d64	5M	1.60	1.58	1.72
regime/mean_jump/context/delta4/d64	209M	1.08	1.10	1.13
tails/fbm/H0.8	5M	1.19	1.25	1.34
tails/fbm/H0.8	209M	1.10	1.13	1.26
in_prior/p24	5M	1.05	1.12	1.35
in_prior/p24	209M	1.03	1.02	1.12

More history helps on Mackey-Glass ($\tau = 17$: 9.09 → 6.47 for 5M and 4.91 → 3.84 for 209M from 2,048 to 32,768 steps) and slightly on Rössler and Lorenz; it does not help on the in-prior control, which gets slightly worse at 32,768 steps. 2 samples (Rössler, c = 5.7) had oracle paths that left the attractor at long horizons; their oracle ratios are excluded.

4.6 Real series that stand in for the families

Real data has no oracle, so it cannot show a gap to the best achievable forecast; it can show whether the families' pattern against classical models carries over. We gathered open-license series (sources, licenses and hashes in `bench/regression/sources.csv` windows whose license does not allow redistribution are kept out of git and rebuilt by `scripts/fetch_realdata.py`). On 412 windows from GIFT-Eval and fev-bench datasets, Timestral's MASE relative to seasonal naive is 0.70 at 5M and 0.65 at 209M (ETS 1.02, ARIMA 1.27; GIFT-Eval-style geometric mean over dataset groups). These datasets may overlap Timestral's pretraining data.

A separate subset of 69 windows was picked as real analogues of the failure families: sunspot numbers, lynx and hormone series and El Niño indices for delay feedback; grid frequency, pedestrian counts and activity recordings for coupled oscillators; river discharge, earthquake counts, financial and crypto series for heavy tails; macro and energy series with documented breaks, traffic and COVID-19 cases for regime switches. The table compares Timestral's ratio to the best classical model on these windows with the same ratio on the synthetic family (context 2,048, horizon 128):

family	real windows	real 5M	synthetic 5M	real 209M	synthetic 209M
delay feedback	12	0.92	0.39	0.73	0.26
coupled oscillators	19	0.79	0.44	0.78	0.20
heavy tails	15	0.90	1.02	0.87	1.01
regime switches	23	1.03	1.02	0.98	0.96

On the synthetic delay and oscillator systems Timestral beats the classical models by a wide margin while staying far from the oracle (Section 4.2); on the real analogues its margin over classical models is smaller. Heavy tails and regime switches sit near 1 on both. With no oracle on the real side, these numbers do not say how far Timestral is from the best achievable forecast there.

5. Covariates and one-way attention

Timestral reads covariates as conditioning rows: the main series reads them, they never read the main series. Several main rows given together read each other both ways. We built six coupled systems with known equations (`otp/generators/coupled.py`) an exogenous driver whose past sets the main series after a lag (LaggedDriver), the same through a nonlinearity (NonlinearDriver), linear and predator-prey feedback loops, a hidden driver seen through a noisy proxy (HiddenDriver), and a known-future covariate that is a noisy copy of the main series shifted by one step (LeakyKnown). The main series is forecast four ways: alone, with the other channel as a past-only covariate, as a known-future covariate where that makes sense, and as a second main row ("joint"). The oracle is matched to the information each condition gives. Sizes 5M and 28M ran on the full grid (5 knob values, 8 seeds, horizons 16 / 128 / 1,024); 123M and 209M on a reduced grid (3 knob values, 4 seeds, horizons 16 / 128), so the oracle gains below differ slightly between the two pairs of sizes. Details and all tables: `results/covariates/README.md`.

Past covariates are used partly, and more with size. When the driver's past determines the main series over the whole horizon, a forecaster that used it fully would cut the error several-fold. Each cell below gives Timestral's covariate gain (MASE alone / MASE with the past covariate) and the matched oracle's gain, at context 2,048 and horizon 16:

system	knob	5M	28M	123M	209M
LaggedDriver	lag 32	1.07 / 8.26	1.66 / 7.86	2.10 / 7.92	2.41 / 7.95
NonlinearDriver	nonlinearity 0.5	1.09 / 6.73	1.62 / 6.48	1.81 / 5.23	2.04 / 4.92
HiddenDriver	x noise 0.05	1.03 / 5.53	1.20 / 5.31	1.78 / 4.35	1.77 / 4.35

At lag 32, 209M reaches a gain of 2.41 out of 7.95 available.

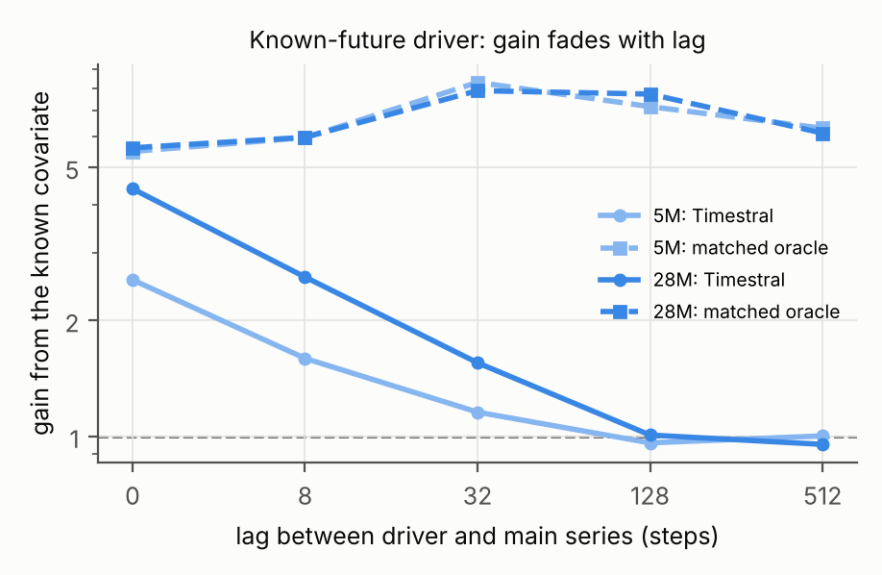


Figure 5: Gain from a known-future driver (MASE alone / MASE with the covariate) on LaggedDriver against the lag, for Timestral (solid) and the oracle that reads the same covariate (dashed); context 2,048, horizon 16, 8 seeds.

Known-future covariates help only when the relation is immediate. With the driver's future given, the 28M gain on LaggedDriver falls from 4.39 at lag 0 to 0.95 at lag 512, while the matched oracle's gain stays at 5.62–7.91 (horizon 16):

lag	5M gain	5M oracle gain	28M gain	28M oracle gain
0	2.55	5.50	4.39	5.62
8	1.59	5.97	2.59	5.98
32	1.15	8.31	1.55	7.91
128	0.96	7.19	1.01	7.75
512	1.00	6.32	0.95	6.13

When the known covariate is the main series itself, one step late with small noise (LeakyKnown at noise ratio 0.05), Timestral's error is several times the oracle's that reads it (MASE over matched-oracle MASE):

size	H=16	H=128	H=1024
5M	8.47	20.92	42.71
28M	3.81	9.94	38.41
123M	3.04	3.70	–
209M	3.25	2.89	–

One-way routing is not the bottleneck. Giving the same channel as a second main row (two-way attention) instead of a past covariate (one-way) changes the error little: joint/past-covariate MASE is within 0.95–1.05 in 95% of the 252 system × knob × size × horizon cells. Among the cells whose 95% interval lies below 1, two-way routing gains most on NonlinearDriver with the nonlinearity off (123M; `results/covariates/summary.csv`), the predator-prey loop (0.81, 95% interval 0.69–0.96) and LaggedDriver (0.83, 0.72–0.92); on the linear feedback loop the ratio stays within 0.98–1.01. The gap to the oracle comes from how much of the covariate's information is used at all, not from which way it is routed.

6. Searching for failures

Sections 4 and 5 test failure families we chose by hand. Here a search procedure chooses. The search space is the mixture-of-primitives generator (`otp/generators/mixture.py`): a genome of numbers in [0, 1] decodes into four component slots (each a trend, season or noise piece, or one of the chaos, regime, heavy-tail or fractal generators, with a weight, a mode of combination and its own parameters) plus an observation-noise level. Every genome is a simulator with an exact Monte Carlo oracle, so every candidate can be scored as in Section 4.

Objective. The fitness of a genome is the geometric mean, over a few fixed seeds, of MASE(Timestral-5M) / MASE(oracle) at context 2,048 and horizon 128 (`otp/search/objective.py`). A genome counts only if the series it makes is forecastable by the oracle: the oracle's sCRPS must be well below seasonal naive's on the same samples (a fixed fraction set in the objective), values must be finite and of sane scale. Infeasible genomes score zero. Every Timestral forward call is counted, and results are cached by genome.

Methods, at equal budget. Each method gets the same number of Timestral calls and three search seeds. (1) CMA-ES over the bounded genome. (2) MAP-Elites, a quality-diversity search: two behaviour descriptors (the dominant component family and a spectral statistic of the context) define a grid of cells, and each cell keeps its best genome, so the archive covers different kinds of failure instead of one peak. (3) Gradient search: a differentiable re-implementation of part of the generator (sinusoids, trend, AR noise with fixed noise draws, a smooth regime gate) is pushed by gradients through Timestral's differentiable forward pass to raise its pinball loss above the oracle's; the final parameters are then scored with the standard pipeline, so all three methods are compared on the same metric. A separate input-space study finds, for in-prior series, the smallest perturbation of the context that doubles Timestral's MASE on a fixed future; it measures sensitivity, not a generator failure, since the oracle is unaffected.

(Results pending: `results/search/comparison.csv` is not in the repository yet.)

Transfer. The top elites of each method are re-scored on 28M, 123M and 209M with fresh seeds, to see whether failures found on the smallest model are failures of the family or of one size.

(Results pending: `results/search/transfer.csv` is not in the repository yet.)

7. Why it fails

7.1 Two suspects that turned out innocent

Symmetric averaging. The released forecaster averages its forecast of the series with the mirrored forecast of the negated series. That forces a symmetric treatment of asymmetric processes (bursty counts, one-sided maps), so it was a natural suspect. Turning it off changes the sCRPS ratio to the oracle by a factor between 0.991 and 1.069 across 14 settings and both small sizes: it is not where the failures come from.

Cumulative normalization. Timestral normalizes every patch with the mean and spread of everything since the start of the context, so old history keeps weighing on the scale after a regime switch. After a variance drop 64 steps back, intervals are indeed far too wide (80% intervals cover at least 0.997 of points; sCRPS ratio 1.59–1.75). But cutting the context down to the post-switch part, which removes the old history from the statistics entirely, lowered the sCRPS ratio in only 12 of 44 cells (those with at least 256 post-switch steps) and raised it in the rest. The anchoring is not the main cause.

7.2 Template or retrieval?

For a deterministic system, the context itself is a library of past trajectories: find the moments whose recent history looks most like the present and see what happened next (the "method of analogues", a few lines of nearest-neighbour code with no training). A second context-only forecaster continues the series as a periodic pattern (the average of its last cycles at the dominant period). On the chaos family (MASE, lower is better):

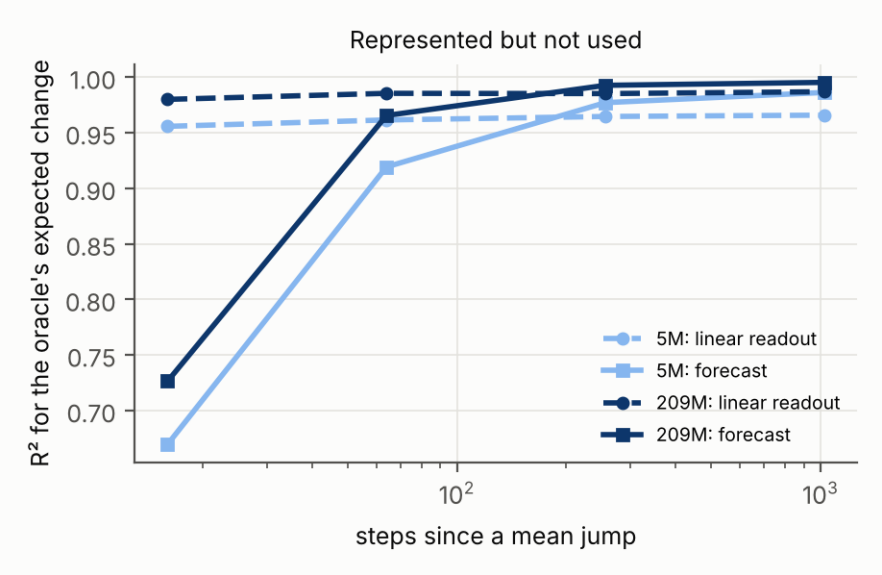


Figure 6: How well the oracle's expected change can be predicted from Timestral's forecast (solid) and from a linear readout of its first forecast token (dashed), against the time since a mean jump.

series	oracle	analog	template	5M	28M	123M	209M
mackey_glass/tau17	0.29	0.93	3.85	2.79	1.84	1.98	1.95
mackey_glass/tau30	0.28	2.20	9.93	6.10	5.96	3.59	3.81
rossler/c5.7	0.48	0.81	2.02	1.59	1.25	1.16	0.98
rossler/c9	0.43	1.25	3.98	3.33	2.71	2.34	1.85
kuramoto/K1	0.20	3.54	3.96	2.61	2.17	2.03	1.94
kuramoto/K8	0.06	0.08	0.87	0.32	0.14	0.12	0.09
lorenz63/rho28	1.44	1.97	2.20	1.89	1.84	1.83	1.79
logistic_map/r3.9	0.54	0.56	0.58	0.56	0.56	0.56	0.56

The method of analogues beats every Timestral size on Mackey-Glass and Rössler and loses to Timestral on weakly coupled Kuramoto and on Lorenz. Timestral's forecasts correlate more with the analogue forecast than with the periodic template, but lean toward the template more than the oracle does (`results/mechanism/analog/summary.csv`). Our reading: Timestral uses the context partly as a trajectory library and partly as a pattern to repeat, and on smooth nonlinear oscillators, which its trend + season + noise prior represents as near-periodic, the second pull wins.

7.3 Inside the model: represented but not used

For each setting we ran one plain forward pass on 300 samples and asked two questions about the oracle's expected change over the first forecast patch (and at single steps 1 and 16): how well does Timestral's own forecast predict it, and how well does a linear readout (a ridge regression, cross-validated) of the hidden state of the first forecast token predict it, at the best layer? If the readout does much better than the output, the information the oracle uses is inside the model but does not reach the forecast. The readout is a lower bound on what is linearly present: with 300 samples and 192–768 features per token, the ridge fit loses some signal, which shows on the in-prior control, where it scores 0.784–0.832 against 0.996–0.997 for the output. Gaps in the readout's favour are therefore conservative.

setting	size	MASE / oracle	output R2	best readout R2	step-1 output R2	step-1 readout R2
mean_jump_d 16	5M	3.624	0.670	0.956	0.540	0.796
mean_jump_d 16	209M	2.476	0.726	0.980	0.584	0.865
mean_jump_d 64	5M	1.536	0.919	0.962	0.803	0.841
mean_jump_d 64	209M	1.121	0.966	0.985	0.905	0.872
mean_jump_d 256	5M	1.087	0.977	0.965	0.899	0.850
mean_jump_d 256	209M	1.026	0.993	0.985	0.951	0.924
freq_jump_d 64	5M	2.269	0.894	0.960	0.084	0.684
freq_jump_d 64	209M	1.856	0.966	0.978	0.227	0.756
freq_jump_d 1024	5M	1.044	0.997	0.961	0.894	0.787
freq_jump_d 1024	209M	1.024	0.997	0.978	0.931	0.890
mackey_tau 17	5M	7.258	0.992	0.999	0.956	0.971
mackey_tau 17	209M	4.855	0.998	1.000	0.984	0.983
lorenz_rho 28	5M	1.415	0.772	0.781	0.724	0.721
lorenz_rho 28	209M	1.346	0.869	0.881	0.972	0.962
in_prior 24	5M	1.070	0.996	0.784	0.958	0.666
in_prior 24	209M	1.027	0.997	0.828	0.989	0.769

Recent regime switches: represented, attended to, not used. Sixteen steps after a mean jump, the readout reaches R^2 0.956–0.980 at every size while the forecast itself reaches 0.570–0.726, and the forecast error is 2.02–3.62 times the oracle's. The forecast tokens are not looking elsewhere: they put 15.70–17.69 times their proportional share of attention on the post-switch patches. Sixty-four steps after a frequency jump, one step ahead, the readout reaches 0.684–0.775 against 0.084–0.227 for the output. By 256 steps after a mean jump the two agree (0.977–0.993 vs. 0.965–0.985). The model sees and encodes a recent change and then forecasts as if it may not last. Our interpretation, which we cannot check without the pretraining data: if lasting breaks are rare in the synthetic trend + season + noise mix, a very recent departure from the pattern is better explained as noise under that prior than as a new regime.

Deterministic oscillators: the gap builds later. On Mackey-Glass ($\tau = 17$) output and readout agree for the first patch (0.992–0.998 vs. 0.999–1.000): the near-term forecast uses what is there, and the error accumulates over the following cycles. There the forecast correlates with a periodic continuation of the context more than the oracle's forecast does (0.87 at 5M, 0.81 at 209M, against 0.751 for the oracle), and the excess shrinks with size together with the error.

8. A neuron atlas

Section 7 looked at a few readouts and attention patterns. This section looks at every MLP neuron of every size. Timestral's MLP is SwiGLU, so a neuron's value (gate times up-projection) takes both signs; each neuron is studied as two poles, one per sign.

Corpus. A labelled corpus of series from the project's generators: in-prior trend, season and noise with controlled parameters, and the chaos, regime, tails and mixture families. For every series the generating variables are stored, and

for every 32-step patch the patch-level variables that can be computed exactly from the simulator: local level, slope and noise, seasonal phase, regime indicator, steps since the last switch, whether the patch holds an outlier, and the oracle's one-step predictability. A seeded split of series is held out and never used for labelling. A mixed-domain set of open-license real windows is added for a transfer test (`results/neurons/real/`).

Labels. For each pole, the known variable (or simple input motif) that best explains its activation on the tokens where that pole is active is found by mutual information and cross-validated linear readout R^2 (single variable and best pair). Each pole stores its best label, score, runner-up and the gap between them.

One job or many. A pole has one main job if its best single label explains at least a fraction θ of the explainable variance, and is unexplained if no label clears a floor. The fractions of one-job, mixed and unexplained poles are reported as curves over θ, per layer and size, never at a single threshold.

size	live poles	one job	95% low	95% high	one job, held-out series	mixed	unexplained
5M	8188	0.045	0.041	0.050	0.041	0.834	0.121
28M	24566	0.029	0.026	0.031	0.024	0.830	0.142
123M	68463	0.027	0.026	0.028	0.022	0.813	0.160
209M	98273	0.023	0.022	0.024	0.019	0.811	0.165

Against this variable library, few poles are single-variable: at $\theta = 0.7$ the one-job share is 4.5% at 5M and 2.3% at 209M, and most poles are mixed (83% and 81%). This number depends on the corpus as much as on the neurons. Scored the same way on corpora of shorter and longer contexts (same generator recipe), the one-job share is 24–47%, and only 12–55% of poles keep the best label they had on the main corpus:

corpus	size	one job	unexplained	mean E	same label as main
main	5M	0.045	0.121	0.257	1.000
main	209M	0.023	0.165	0.212	1.000
short	5M	0.415	0.240	0.209	0.212
short	209M	0.308	0.457	0.136	0.234
long	5M	0.242	0.314	0.189	0.551
long	209M	0.242	0.418	0.153	0.547
xlong	5M	0.475	0.380	0.164	0.198
xlong	209M	0.431	0.444	0.146	0.122

The main corpus has the largest explainable ceiling E (more of each pole's variance is explained by combinations of variables), which lowers every single-variable ratio. A control separates the possible causes (`results/neurons/followup/`). On the univariate part of the main corpus (5M: 4.8% one-job with the full library), restricting the library to the variables defined in all four corpora gives 4.8%, and random libraries of 10 to 50 variables give 3.2–7.2%. Subsampling the main corpus to fewer series raises it: 11–12% with 1,034 series and 26–37% with 48, with E falling and the labels drifting away from the full-corpus ones:

size	corpus	series	library	runs	one job (mean)	min	max	mean E	same label as reference
5M	long	620	common	1	0.226	0.226	0.226	0.193	0.516
5M	main	3100	common	1	0.048	0.048	0.048	0.262	0.989
5M	main	3100	full	1	0.048	0.048	0.048	0.262	1.000
5M	main	1034	full	3	0.115	0.111	0.122	0.214	0.788
5M	main	620	full	3	0.264	0.262	0.266	0.156	0.731

size	corpus	series	library	runs	one job (mean)	min	max	mean E	same label as reference
5M	main	48	full	3	0.326	0.257	0.374	0.124	0.297
5M	short	1034	common	1	0.408	0.408	0.408	0.211	0.194
5M	xlong	48	common	1	0.474	0.474	0.474	0.164	0.172
28M	main	3100	common	1	0.032	0.032	0.032	0.232	0.982
28M	main	3100	full	1	0.032	0.032	0.032	0.233	1.000
28M	main	1034	full	1	0.103	0.103	0.103	0.182	0.768
28M	main	48	full	1	0.303	0.303	0.303	0.116	0.231

So the share is driven mostly by how many series the scores are estimated from: with fewer series the cross-validated ceiling E shrinks and single-variable ratios rise. The main-corpus value is the best estimate we have, and it was still falling between 1,034 and 3,100 series, so it is an upper bound on the share a larger corpus would give. We report it as "against known generating variables, few poles are single-variable", and do not report it as evidence that the neurons are polysemantic in general. A label space that is not fixed in advance (for example motif shapes named from the exemplars, as interpretability work on language models does) could give a different answer.

Checks. Predictive: on the held-out series, each pole's activation is predicted from its label alone and the held-out R^2 is compared with the labelling-set score. Causal: for a stratified sample of poles, the pole is scaled during the forward pass and the change in the forecast is compared with a prediction written down, and committed, before any steering run (`results/neurons/steering_preregistration.md`).

size	poles	n	matched	match rate	95% high	monotone share
123M	labeled	68	0	0.000	0.053	0.574
123M	complement	26	0	0.000	0.129	0.500
123M	control	432	0	0.000	0.009	0.486
209M	labeled	63	0	0.000	0.057	0.603
209M	complement	20	0	0.000	0.161	0.650
209M	control	432	0	0.000	0.009	0.521
28M	labeled	47	0	0.000	0.076	0.532
28M	complement	20	0	0.000	0.161	0.600
28M	control	432	0	0.000	0.009	0.481
5M	labeled	25	0	0.000	0.133	0.320
5M	complement	11	0	0.000	0.259	0.364
5M	control	432	5	0.012	0.027	0.449

No labeled pole matched its prediction. This does not show that the labels are not causal: scaling a single pole (up to four times, or to zero) moved the forecast by at most 0.041 context standard deviations over all 203 steered labeled poles, below every pre-registered size threshold, so the test had no power to detect a match. A test with power needs a positive control (steering a direction the model is known to use, scaled until the measured effect is clear) and group steering; that follow-up is described in `results/neurons/README.md` when it has run.

Link to the failure map. Which poles are active on the Mackey-Glass, Rössler and Kuramoto contexts where Timestral falls behind the oracle, whether seasonal or periodic-pattern poles are among them, and whether damping them moves the forecast toward the oracle or toward the analogue forecaster of Section 7.2.

size	chaos-label enrichment among elevated poles	MASE ratio before	after ablating periodic poles	after ablating random poles
5M	2.487	1.850	1.850	1.851

size	chaos-label enrichment among elevated poles	MASE ratio before	after ablating periodic poles	after ablating random poles
28M	2.072	1.754	1.759	1.755
123M	1.957	1.685	1.688	1.683
209M	2.017	2.050	2.050	2.053

Poles with chaos-related labels are over-represented among the poles that are more active on failing chaos contexts than on reference contexts, at every size. Ablating the elevated poles with periodic labels leaves the forecast's error ratio essentially where it was, as does ablating as many random poles in the same layers: the periodic-template behaviour of Section 7.2 is not carried by a small set of neurons that can be switched off.

9. Benchmarks

The failure map is released as two benchmark sets built from the same simulators (code and data in `bench/`). Each series comes with the oracle's 21 quantiles, so a model is scored against the best achievable forecast as well as against seasonal naive, the GIFT-Eval convention.

OffPrior-Diag has 372 series: 31 configs with 12 seeds each, at context 2,048 and horizon 128. Every config was picked to cover one failure type from Sections 4 and 7, or to serve as a control (inside the prior, an old switch, a switch inside the horizon that nobody can see coming, a map nobody can forecast); each series carries its failure type and a one-line reason. **OffPrior-LB** has 3,776 series: every config of the chaos, regime, tails and in-prior grids, 40 fixed random mixture genomes and four more in-prior periods (236 configs), 16 seeds per config spread over contexts 512 / 2,048 / 8,192 and horizons 16 / 128 / 1,024, split into public and held-out halves by seed parity. The seeds and code are public, so the held-out half is a convenience, not a secret test set. `bench/eval.py` scores one model in one command; the summary is the geometric mean over series of MASE and sCRPS divided by seasonal naive's, and of MASE divided by the oracle's.

model	series	MASE / seasonal naive	sCRPS / seasonal naive	MASE / oracle	80% coverage
oracle	372	0.272	0.209	1.000	0.789
timestral-209M	372	0.543	0.427	2.002	0.781
timestral-123M	372	0.550	0.435	2.028	0.771
timestral-28M	372	0.607	0.478	2.239	0.751
timestral-5M	372	0.656	0.515	2.417	0.705
ets	372	1.213	1.190	4.468	0.875
arima	372	1.123	0.872	4.150	0.738
seasonal_naive	372	1.000	1.000	3.679	0.814

model	series	MASE / seasonal naive	sCRPS / seasonal naive	MASE / oracle	80% coverage
oracle	3776	0.463	0.286	1.000	0.798
timestral-209M	3776	0.608	0.385	1.314	0.793
timestral-123M	3776	0.615	0.389	1.329	0.791
timestral-28M	3776	0.637	0.402	1.375	0.783
timestral-5M	3776	0.669	0.425	1.446	0.779
ets	3776	1.006	0.926	2.172	0.889
arima	3776	1.039	0.663	2.245	0.770
seasonal_naive	3776	1.000	1.000	2.159	0.882

On OffPrior-Diag, Timestral's MASE is 2.00× the oracle's at 209M and 2.42× at 5M, against 4.47× for ETS, 4.15× for ARIMA and 3.68× for seasonal naive. On OffPrior-LB the same ratios are 1.31× and 1.45× for Timestral, and 2.17×, 2.24× and 2.16× for the classical models. By family on OffPrior-LB, 209M is at 2.16× the oracle on chaos, 1.47× on the mixture genomes and 1.01× on tails.

failure type	series	209M MASE / oracle	209M MASE / seasonal naive
bursty	12	0.991	0.938
chaos_short_reach	24	4.455	0.699
control_in_prior	60	1.007	0.699
control_old_switch	12	1.016	0.745
control_unknowable	12	0.978	0.804
control_unpredictable	12	1.033	0.764
coupled_oscillators	36	7.942	0.177
delay_oscillator	36	6.515	0.223
heavy_tail_walk	12	1.006	1.011
long_memory	12	1.123	0.643
multifractal	12	1.033	1.037
periodic_map	12	1.187	0.046
recent_switch	84	1.808	1.074
smooth_chaos	24	2.875	0.273
switching	12	1.090	0.753

By failure type, the diagnostic set reproduces the map at 209M: coupled oscillators (7.94×), delay oscillators (6.52×), chaos within the first Lyapunov times (4.46×), smooth chaos (2.87×) and recent switches (1.81×), while the controls and the heavy-tailed, bursty and multifractal types stay at 0.98–1.03×.

The published files were built on a Windows laptop. A rebuild on Linux with newer NumPy matched them array-for-array for 25 of 31 diagnostic configs; in the others (Mackey-Glass, Kuramoto near synchronization, and one fractional Brownian motion setting) platform floating-point differences change the simulated paths, and the chaotic ones diverge. Scores should therefore be reported on the published files (`bench/data/cross_platform_diag.json` has the per-config comparison).

10. Fixing it

10.1 Without training: let the context decide

Section 7.2 showed that a nearest-neighbour forecaster built from the context beats Timestral on some deterministic systems and loses badly on others. A forecaster can find out which case it is in from the context alone: forecast the last stretch of the context from what comes before it, with both methods, and compare. The rule we test switches to the analogue forecast (keeping Timestral's quantile spread around the new median) only when the analogues' backtest error is below 0.7 times Timestral's on each of the last two 128-step windows.

How the rule was chosen matters. A first pre-set procedure (tune on even seeds, report on odd seeds) picked a one-window rule that helped chaos but cost the tails family 1.031–1.043 on the held-out seeds. We then picked the two-window rule from that held-out table, which is a choice made after seeing it, so we froze it and evaluated it once more on fresh samples generated for that purpose (`results/fix/README.md` lists all three stages). On the fresh samples (MASE of the fix divided by Timestral's MASE; below 1 is better):

family	5M	28M	123M	209M	share switched (range over sizes)
chaos	0.824	0.877	0.970	0.982	0.067-0.238
regime	1.000	1.000	1.000	1.000	0.000-0.000

family	5M	28M	123M	209M	share switched (range over sizes)
tails	1.003	1.000	1.001	1.000	0.004-0.011
in_prior	1.000	1.000	1.000	1.000	0.000-0.000

The fix improves chaos by a factor of 0.824 at 5M and 0.982 at 209M, leaves in-prior and regime series untouched (the rule never switches there), and costs at most a factor of 1.003 on tails. Its benefit shrinks with model size, in line with larger models closing more of the chaos gap on their own.

The other zero-training idea, cutting the context to the part after a detected change (Section 7.1), helps only when at least 256 post-switch steps are available.

10.2 Fine-tuning: full, LoRA, and data mixing

The direct fix is to train on the failures. We compare full fine-tuning (every weight), LoRA (low-rank adapters on the attention and MLP projections, base weights frozen) and data mixing (a share of every batch replaced by in-prior trend, season and noise series, on top of either method), with training code in `otp/finetune.py` (pinball loss over the 21 quantiles, scaled by the context's naive error; AdamW with warmup and cosine decay). Two held-out protocols keep the test away from the training distribution: leave-one-family-out (train on three of chaos, regime, tails and mixture, test on the fourth) and held-out knob ranges (train on one range of each family's knob, test on the other; splits in `otp/finetune_splits.py`). A regression check scores the fine-tuned models on normal series (the in-prior set and, where available, the real-data set) against the pretrained model of the same size. Gain is reported as the paired MASE ratio fine-tuned / pretrained on the held-out samples (500-path oracle), with 95% intervals, next to cost (trainable parameters, steps, series seen, wall time).

A first pass on the 5M model ran on CPU with a short budget, to validate the pipeline end to end:

(Results pending: `results/finetune/local/summary.csv` is not in the repository yet.)

(Results pending: `results/finetune/local/summary.csv` is not in the repository yet.)

(Results pending: `results/finetune/local/summary.csv` is not in the repository yet.)

The full-budget runs on all four sizes ran on GPUs:

(Results pending: `results/finetune/colab/summary.csv` is not in the repository yet.)

11. Limitations

- **The oracle knows the latent state.** It is the best forecast given the true state, which is stronger than the best forecast given only the observed history. Ratios to it therefore bound the possible improvement from above; part of every gap is unavoidable for any forecaster that sees only the data.
- **Noise sets the size of chaos gaps.** For nearly deterministic systems the oracle's error, and so every ratio to it, depends on the noise level we chose (1% by default); Section 4.2 reports the dependence. Orderings (between settings and between model sizes) are more robust than the magnitudes.
- **Seeds.** Most settings use 8 seeds (16 for the noise sweep, 300 samples per setting for the mechanism runs). Intervals in the result files are bootstrap intervals over seeds; with 8 seeds they are wide for heavy-tailed settings.
- **Diverged simulations.** A few Rössler simulations left the attractor under noise; their oracle could not be scored and they are excluded (counts are given where they occur).
- **Univariate main series.** Except for the covariate study, every test series is a single variate.
- **No access to the pretraining data.** Statements about what the synthetic prior contains are inferences from behaviour, labelled as such.
- **Compute.** Small-model sweeps ran on a 2-core CPU VM; the large-model sweeps, the long-context runs and the mechanism runs at all sizes ran on Colab A100 GPUs; one mechanism test ran on a laptop RTX 3060 GPU. GPU and CPU forecasts of the same series agree to a few parts in 100,000 in MASE (`results/mechanism/analog/parity_vm_cpu_vs`

Reproducibility

All code, generator seeds and result files are in the repository. Every number above points to the result file it comes from; every result file has a `provenance.json` naming the script, the commit, the host and the wall time.